

Theory of Automata/Computation and Formal Languages Notes

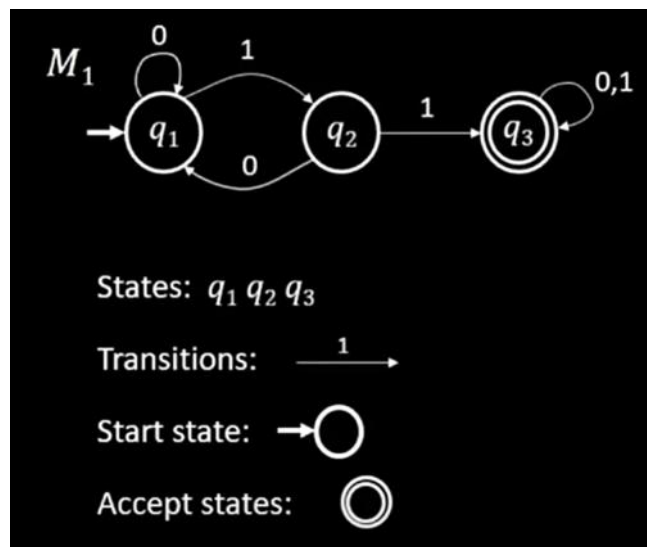
The Solitude Spot

Monday, 8 September 2025 7:11 pm

WEEK 0 - LECTURE 0

FINITE AUTOMATON:

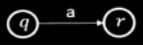
- ❖ Abstract models of computers that are much simpler to understand and capture the important aspects of things we are trying to understand
- ❖ A machine with a small limited amount of memory (think of it this way)



- ❖ For the state diagram above our 5 tuple $(Q, \Sigma, \delta, q_0, F)$ would be
- ❖ $Q = \{q_1, q_2, q_3\}$
- ❖ $\Sigma = \{0, 1\}$
- ❖ $q_0 = q_1$
- ❖ $F = q_3$

δ	0	1
q_1	q_1	q_2
q_2	q_1	q_3
q_3	q_3	q_3

- ❖ Giving it a thought, substrings that are composed of at least 2 consecutive 1's take us to the acceptance state
- ❖ Strings that get accepted are the language of our machine or that M1 recognizes $A / L(M1)$
- ❖ In this case $A = L(M1)$, where A contains the substring 11
- ❖ The language recognized by the machine is the collection of the strings the machine accepts
- ❖ The machine always recognizes one language even when it accepts no strings, the automata is recognizing the empty language

- ❖ A finite automaton is a 5 tuple $(Q, \Sigma, \delta, q_0, F)$
- ❖ Q is finite the total number of states
- ❖ Σ is the finite set of alphabet symbols
- ❖ δ represents the function that is $\delta = Q \times \Sigma$
 $\delta(q, a) = r$
- ❖ Delta is the function that tells us the transition of our machine, which takes in a state and an alphabet to give us our next state
- $\delta: Q \times \Sigma \rightarrow Q$
 $\delta(q, a) = r$ means 
- ❖ q_0 is the initial state (only 1)
- ❖ F is the final/acceptance state which can be more than 1 or zero

- ❖ Input is going to be some finite string
- ❖ Output will either be an acceptance or a rejection
- ❖ Computation will be beginning at the start state (q_1), follow through with transitions by reading the input string symbol by symbol
- ❖ If input string ends at the final/acceptance state it is accepted otherwise rejected

Example:

- ❖ 01101 receives an acceptance
- ❖ 00101 gets rejected

Language:

- ❖ Set of strings (maybe finite or infinite)

String:

- ❖ A finite sequence of symbols in Σ

Special String:

- ❖ ϵ is an empty string with a length of zero

Special Language:

- ❖ $\Phi / \{\}$ is an empty language set with no strings
- ❖ Φ has the cardinality/length of zero and not 1, the language with an empty string only has a length of 1

Regular Language:

- ❖ A language is regular if some finite automaton recognizes it
- ❖ From the above example $A / L(M1)$ is a regular language

Irregular Language:

- ❖ A language for which there is no way to get that language recognized with a finite automaton

REGULAR EXPRESSION

Regular Operations:

- ❖ Let A and B be two languages
- ❖ Where $A = \{\text{abu, bakar}\}$ and $B = \{\text{good, bad}\}$

Binary Operations:

- ❖ Union: combines/joins all the string from the two languages
 $A \cup B = \{\text{abu, bakar, good, bad}\}$
- ❖ Concatenation: takes a strings from language A and multiplies that with every string from the B language
- ❖ We get all possible ways of having a string from the A language followed by all possible ways of having a string from the B language
 AB or $A \circ B = \{\text{abugood, abubad, bakargood, bakarbad}\}$

Unary Operations:

- ❖ Star Operation: gives a language that can have all the and any number of members from the original language
- ❖ Basically it takes a bunch of (any number of) strings from the original language and just join them together, which becomes an element of the language after the star operation
- ❖ It always gives us an infinite language if A (the original language) contains any non-empty member
- ❖ This also allows us to stick zero elements together so ϵ is always a member of the language after this operation is performed
 $A^* = \{\epsilon, \text{abu, bakar, abuabu, abubakar, bakarabu, bakarbakar, abuabuabu, abuabubakar,}\}$
 $\Phi^* = \{\epsilon\}$ empty language star = a language with just the empty string
- ❖ $\Sigma^* = (0 \cup 1)^*$ given that $\Sigma = \{0, 1\}$
- ❖ $(0 \cup 1)^* 1 = \Sigma^* 1 =$ all strings that end with a 1
- ❖ $(0 \cup 1)^* 11 (0 \cup 1)^* =$ this is indeed the regular expression for the finite automaton in the diagram above, (all string that have 11 as a substring)

Closure Properties:

- ❖ Theorem - Closure under $\cup$: if A_1 and A_2 are regular languages then so is $A_1 \cup A_2$
- ❖ Theorem - Closure under $\circ$: if A_1 and A_2 are regular languages then so is $A_1 \circ A_2$